PROJECT REPORT

Encryption & Decryption using Caesar Cipher

DecodeLabs Cyber Security Internship – Batch 2026

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Project Number: 2

Project Title: Encryption & Decryption using Caesar Cipher

Abstract

This project implements a Caesar Cipher-based encryption and decryption system using Python and Tkinter. The application allows users to transform plaintext into ciphertext using a configurable shift key and decrypt the ciphertext back into its original form. The project demonstrates fundamental cryptographic concepts, character manipulation techniques, and graphical user interface development.

The application features a modern cybersecurity-inspired interface, encryption and decryption functionality, customizable shift values, real-time output display, clipboard support, and round-trip verification. The project serves as an introduction to cryptography and secure data transformation techniques used in cybersecurity.

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1. Introduction

Cryptography is the science of protecting information through mathematical transformations. Encryption converts readable data into an unreadable format, while decryption restores the original information.

This project demonstrates the implementation of the Caesar Cipher, one of the earliest and simplest encryption techniques. The application provides a graphical interface for performing encryption and decryption operations while helping users understand the underlying cryptographic process.

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2. Objective

- To understand basic cryptographic concepts.
- To implement Caesar Cipher encryption and decryption.
- To develop a graphical user interface using Python Tkinter.
- To provide a user-friendly encryption simulator.
- To understand data confidentiality and secure communication principles.

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3. Problem Statement

Sensitive information transmitted over communication channels can be intercepted by unauthorized users. Encryption provides a mechanism for protecting such information by converting it into a form that cannot be easily understood without the appropriate key.

The objective is to build an application capable of encrypting and decrypting messages using a shift-based substitution cipher.

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4. Technology Stack

Component	Technology
Language	Python
GUI Framework	Tkinter

Component	Technology
Algorithm	Caesar Cipher
Clipboard Support	Pyperclip
Platform	Windows

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5. Algorithm Used

Caesar Cipher

The Caesar Cipher shifts each alphabetic character by a fixed number of positions.

Encryption Formula

$E(x) = (x + n) \mod 26$

Where:

- x = Character position
- n = Shift value

Decryption Formula

$D(x) = (x - n) \mod 26$

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6. System Workflow

Plaintext Input

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Encryption Process

↓

Ciphertext Output

↓

Decryption Process

↓

Original Plaintext

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7. GUI Design

Figure 1 – Application Launch Screen

The application starts with a cybersecurity-themed graphical interface containing:

- Plaintext input area
- Shift key configuration
- Encryption controls
- Output display panels
- Clipboard utility buttons

Figure 2 – Encryption Operation

The user enters plaintext and specifies a shift key. The system generates the corresponding ciphertext and displays encryption statistics.

Figure 3 – Decryption Verification

The encrypted text is decrypted using the same shift key. The application confirms successful round-trip conversion back to the original message.

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8. Features Implemented

Core Features

- Caesar Cipher Encryption
- Caesar Cipher Decryption
- Custom Shift Key Support
- Round-Trip Verification

GUI Features

- Dark Cybersecurity Theme
- Input Text Area
- Encrypted Output Panel
- Decrypted Output Panel
- Copy Encrypted Text
- Copy Decrypted Text

- Clear Functionality

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9. Test Cases and Results

Test Case 1

Input:

Hello@123

Shift Key:

3

Encrypted Output:

Khoor@123

Decrypted Output:

Hello@123

Result:

Pass

Test Case 2

Input:

Cyber Security

Shift Key:

3

Encrypted Output:

Fbehu Vhfxulwb

Result:

Pass

Test Case 3

Input:

XYZ

Shift Key:

3

Encrypted Output:

ABC

Result:

Pass

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10. Advantages

- Easy to understand and implement.
- Demonstrates basic cryptographic concepts.
- Supports custom shift values.
- Handles letters, numbers, spaces, and symbols.
- User-friendly graphical interface.

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11. Limitations

- Small key space.
- Vulnerable to brute-force attacks.
- Not suitable for modern secure communication.
- Intended for educational purposes.

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12. Learning Outcomes

Through this project, the following concepts were learned:

- Fundamentals of cryptography.
- Encryption and decryption techniques.
- ASCII character manipulation.
- Python GUI development.
- Event-driven programming.
- Secure data handling concepts.

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13. Conclusion

The Encryption & Decryption using Caesar Cipher project successfully demonstrates the fundamental principles of cryptography. The developed application encrypts plaintext into ciphertext and accurately restores the original text through decryption. The project provides practical experience in implementing encryption algorithms and graphical user interface design using Python.

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14. Future Enhancements

- AES Encryption
- ROT13 Support
- Base64 Encoding and Decoding
- File Encryption
- Password-Based Encryption
- Encryption History Dashboard
- Multi-Algorithm Encryption Suite

Result

The project was successfully designed, implemented, tested, and verified. All encryption and decryption operations produced the expected results.